THE UK UNIVERSITY INTEGRATION BEE 2024/2025

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ROUND THREE TOURNAMENT SOLUTIONS

Sponsored by


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1 Battle 1

1.1 Fight 1

1. $\displaystyle\int_0^\infty \frac{\lfloor x \rfloor}{1+\lfloor x \rfloor^2+\lfloor x \rfloor^4}\,\mathrm{d}x$

Solution. Telescoping by partial fractions gives:

$$\begin{aligned}\int_0^\infty \frac{\lfloor x \rfloor}{1+\lfloor x \rfloor^2+\lfloor x \rfloor^4}\,\mathrm{d}x &= \sum_{n=0}^{\infty} \frac{n}{1+n^2+n^4} \\ &= \frac{1}{2}\sum_{n=0}^{\infty}\left(\frac{1}{(n-1)n+1}-\frac{1}{n(n+1)+1}\right) \\ &= \frac{1}{2}.\end{aligned}$$

:)

2. $\displaystyle\int_0^{\frac{\pi}{4}} (4\cos^2(x)-1)\sqrt{\tan(x)}\,\mathrm{d}x$

Solution. We have

$$4\cos^2(x)-1 = 2\cos(2x)+1.$$

Now we perform a partial IBP. We have

$$\begin{aligned}\int_0^{\frac{\pi}{4}} 2\cos(2x)\sqrt{\tan(x)}\,\mathrm{d}x &= \left[\sin(2x)\sqrt{\tan(x)}\right]_0^{\frac{\pi}{4}} - \int_0^{\frac{\pi}{4}} \sin(2x)\frac{\sec^2(x)}{2\sqrt{\tan(x)}}\,\mathrm{d}x \\ &= 1-\int_0^1 \sqrt{\tan(x)}\,\mathrm{d}x.\end{aligned}$$

Hence it follows that

$$\int_0^{\frac{\pi}{4}} (4\cos^2(x)-1)\sqrt{\tan(x)}\,\mathrm{d}x = 1.$$

:)

3. $\displaystyle\int_0^{\frac{\pi}{2}} \frac{2-\sqrt{\tan(x)}+\tan(x)}{1+\tan^{3/2}(x)}\,\mathrm{d}x$

Solution. King's in disguise. Since $\tan\left(\frac{\pi}{2}-x\right)=\cot(x)$, we have

$$\begin{aligned}I &:= \int_0^{\frac{\pi}{2}} \frac{2-\sqrt{\tan(x)}+\tan(x)}{1+\tan^{3/2}(x)}\,\mathrm{d}x \\ &= \int_0^{\frac{\pi}{2}} \frac{2-\sqrt{\cot(x)}+\cot(x)}{1+\cot^{3/2}(x)}\,\mathrm{d}x \\ &= \int_0^{\frac{\pi}{2}} \frac{2\tan^{3/2}(x)-\tan(x)+\sqrt{\tan(x)}}{1+\tan^{3/2}(x)}\,\mathrm{d}x.\end{aligned}$$

Then we get

$$I = \frac{1}{2}\int_0^{\frac{\pi}{2}} \frac{2+2\tan^{3/2}(x)}{1+\tan^{3/2}(x)}\,\mathrm{d}x = \frac{\pi}{2}.$$

:)

1.2 Fight 2

1. $\displaystyle\int \frac{x-1}{(x-2)(x-3)} + \frac{x-2}{(x-3)(x-1)} + \frac{x-3}{(x-1)(x-2)}\,\mathrm{d}x$

   Solution. An exercise in how fast can you do partial fractions. We have

   $$\begin{aligned}&\int \frac{x-1}{(x-2)(x-3)} + \frac{x-2}{(x-3)(x-1)} + \frac{x-3}{(x-1)(x-2)}\,\mathrm{d}x\\ &= \frac{5}{2}\int \frac{1}{x-3}\,\mathrm{d}x + \frac{5}{2}\int \frac{1}{x-1}\,\mathrm{d}x - 2\int \frac{1}{x-2}\,\mathrm{d}x\\ &= \frac{5}{2}\ln|x-3| + \frac{5}{2}\ln|x-1| - 2\ln|x-2| + C.\end{aligned}$$

   :)

2. $\displaystyle\int_1^\infty \frac{1}{x(x^{\ln(2048)}+1)}\,\mathrm{d}x$

   Solution. The trick is to factor the denominator.

   $$\int_1^\infty \frac{1}{x(x^{\ln(2048)}+1)}\,\mathrm{d}x = \int_1^\infty \frac{dx}{x^{\ln(2048)+1}(1+x^{-\ln(2048)})}\,\mathrm{d}x$$

   Let $u = 1 + x^{-\ln(2048)}$. Then $u' = -\ln(2048)x^{-\ln(2048)-1}$, and hence

   $$\int_1^\infty \frac{dx}{x^{\ln(2048)+1}(1+x^{-\ln(2048)})}\,\mathrm{d}x = \frac{1}{\ln(2048)}\int_1^2 \frac{1}{u}\,\mathrm{d}u = \frac{\ln(2)}{11\ln(2)} = \frac{1}{11}.$$

   :)

3. $\displaystyle\int_0^{2\pi} \frac{1}{(1+2^{\sin(x)})(1+2^{\cos(x)})}\,\mathrm{d}x$

   Solution. By King's rule, we have

   $$\begin{aligned}I &:= \int_0^{2\pi} \frac{1}{(1+2^{\sin(x)})(1+2^{\cos(x)})}\,\mathrm{d}x\\ &= \int_0^{2\pi} \frac{1}{(1+2^{\sin(2\pi-x)})(1+2^{\cos(2\pi-x)})}\,\mathrm{d}x\\ &= \int_0^{2\pi} \frac{1}{(1+2^{-\sin(x)})(1+2^{\cos(x)})}\,\mathrm{d}x\\ &= \int_0^{2\pi} \frac{2^{\sin(x)}}{(1+2^{\sin(x)})(1+2^{\cos(x)})}\,\mathrm{d}x\end{aligned}$$

Hence we get, by way of symmetry and shift properties,

$$\begin{aligned}
I &= \frac{1}{2}\int_0^{2\pi} \frac{1}{1+2^{\cos(x)}}\,\mathrm{d}x \\
&= \frac{1}{2}\left(\int_0^{\pi} \frac{1}{1+2^{\cos(x)}}\,\mathrm{d}x + \int_{\pi}^{2\pi} \frac{1}{1+2^{\cos(x)}}\,\mathrm{d}x\right) \\
&= \frac{1}{2}\left(\int_0^{\pi} \frac{1}{1+2^{\cos(x)}}\,\mathrm{d}x + \int_0^{\pi} \frac{1}{1+2^{\cos(x+\pi)}}\,\mathrm{d}x\right) \\
&= \frac{1}{2}\left(\int_0^{\pi} \frac{1}{1+2^{\cos(x)}}\,\mathrm{d}x + \int_0^{\pi} \frac{1}{1+2^{-\cos(x)}}\,\mathrm{d}x\right) \\
&= \frac{1}{2}\int_0^{\pi} \frac{1+2^{-\cos(x)}+1+2^{\cos(x)}}{(1+2^{\cos(x)})(1+2^{-\cos(x)})}\,\mathrm{d}x \\
&= \frac{\pi}{2}.
\end{aligned}$$

:)

1.3 Fight 3

1. $\displaystyle\int_{-1}^{1} \frac{2025+e^x+e^{-x}}{1+2025^x}\,\mathrm{d}x$

   Solution. An application of the identity

   $$\int_{-a}^{a} \frac{e(x)}{1+b^{o(x)}}\,\mathrm{d}x = \int_0^a e(x)\,\mathrm{d}x,$$

   where $e(x)$ is an even function and $o(x)$ is an odd function. With $e(x) = 2025 + e^x + e^{-x}$ and $o(x) = x$, we get

   $$\begin{aligned}
   \int_{-1}^{1} \frac{2025+e^x+e^{-x}}{1+2025^x}\,\mathrm{d}x &= \int_0^1 2025+e^x+e^{-x}\,\mathrm{d}x \\
   &= [2025x + e^x - e^{-x}]_0^1 \\
   &= 2025 + e - \frac{1}{e}.
   \end{aligned}$$

   :)

2. $\displaystyle\int_0^{\infty} \frac{1}{\sinh(\sqrt{x})+\operatorname{csch}(\sqrt{x})}\,\mathrm{d}x$

   Solution. By using hyperbolic identities,

   $$\begin{aligned}
   \int_0^{\infty} \frac{1}{\sinh(\sqrt{x})+\operatorname{csch}(\sqrt{x})}\,\mathrm{d}x &= \int_0^{\infty} \frac{\sinh(\sqrt{x})}{\sinh^2(\sqrt{x})+1}\,\mathrm{d}x \\
   &= \int_0^{\infty} \tanh(\sqrt{x})\operatorname{sech}(\sqrt{x})\,\mathrm{d}x \\
   &= 2\int_0^{\infty} u\tanh(u)\operatorname{sech}(u)\,\mathrm{d}u \\
   &= 2[-u\operatorname{sech}(u)]_0^{\infty} + 2\int_0^{\infty} \operatorname{sech}(u)\,\mathrm{d}u \\
   &= 2[\tan^{-1}(\sinh(u))]_0^{\infty} \\
   &= \pi.
   \end{aligned}$$

   :)

3. $\displaystyle\int_0^1 \sqrt{x+x^3}+\sqrt{x+\sqrt[3]{x}}\,\mathrm{d}x$

   Solution. On the second term, let $u^3 = x$. Then we have

$$\int_0^1 \sqrt{x+\sqrt[3]{x}}\,\mathrm{d}x = \int_0^1 3u^2\sqrt{u^3+u}\,\mathrm{d}u.$$

   Then it follows

$$\int_0^1 \sqrt{x+x^3}+\sqrt{x+\sqrt[3]{x}}\,\mathrm{d}x = \int_0^1 (3x^2+1)\sqrt{x+x^3}\,\mathrm{d}x = \left[\frac{2}{3}(x+x^3)^{\frac{3}{2}}\right]_0^1 = \frac{4\sqrt{2}}{3}.$$

:)

1.4 Tiebreaker

1. $\displaystyle\int_0^{\frac{\pi}{2}} \frac{\sin(4x)}{\sin(x)}\,\mathrm{d}x$

   Solution. Just apply double angle identities:

$$\begin{aligned}\int_0^{\frac{\pi}{2}} \frac{\sin(4x)}{\sin(x)}\,\mathrm{d}x &= \int_0^{\frac{\pi}{2}} \frac{4\sin(x)\cos(x)(1-2\sin^2(x))}{\sin(x)}\,\mathrm{d}x \\ &= \int_0^{\frac{\pi}{2}} 4\cos(x)-8\cos(x)\sin^2(x)\,\mathrm{d}x \\ &= \left[4\sin(x)-\frac{8}{3}\sin^3(x)\right]_0^{\frac{\pi}{2}} \\ &= \frac{4}{3}.\end{aligned}$$

:)

2 Battle 2

2.1 Fight 1

1. $\displaystyle\int_1^e \frac{x\ln(x)}{(1+\ln(x))^3}\,\mathrm{d}x$

   Solution. Put $u = 1+\ln(x)$ to get

$$\begin{aligned}\int_1^e \frac{x\ln(x)}{(1+\ln(x))^3}\,\mathrm{d}x &= \frac{1}{e^2}\int_1^2 \frac{e^{2u}(u-1)}{u^3}\,\mathrm{d}u\\ &= \frac{1}{e^2}\int_1^2 \frac{e^{2u}(u^2-u)}{u^4}\,\mathrm{d}u\\ &= \frac{1}{2e^2}\left[\frac{e^{2u}}{u^2}\right]_1^2\\ &= \frac{1}{2}\left(\frac{e^2}{4}-1\right).\end{aligned}$$

   :)

2. $\displaystyle\int_0^\infty \frac{\tan^{-1}(x)}{x((\ln(x))^2+1)}\,\mathrm{d}x$

   Solution. Perform a border flip $x = \dfrac{1}{y}$ to get

$$\begin{aligned}I &:= \int_0^\infty \frac{\tan^{-1}(x)}{x((\ln(x))^2+1)}\,\mathrm{d}x\\ &= \int_0^\infty \frac{y\tan^{-1}\left(\frac{1}{y}\right)}{\left(\ln\left(\frac{1}{y}\right)\right)^2+1}\cdot\frac{1}{y^2}\,\mathrm{d}y\\ &= \int_0^\infty \frac{\tan^{-1}\left(\frac{1}{y}\right)}{y((\ln(y))^2+1)}\,\mathrm{d}y.\end{aligned}$$

   Thus we have

$$\begin{aligned}I &= \frac{1}{2}\int_0^\infty \frac{\tan^{-1}(x)+\tan^{-1}\left(\frac{1}{x}\right)}{x((\ln(x))^2+1)}\,\mathrm{d}x\\ &= \frac{\pi}{4}\int_0^\infty \frac{\frac{1}{x}}{(\ln(x))^2+1}\,\mathrm{d}x\\ &= \frac{\pi}{4}[\tan^{-1}(\ln(x))]_0^\infty\\ &= \frac{\pi^2}{4}.\end{aligned}$$

   :)

3. $\displaystyle\int_1^\infty \frac{1}{x\sqrt{\sqrt[e]{x}-1}}\,\mathrm{d}x$

   Solution. Let $u = \sqrt{\sqrt[e]{x}-1}$. Then $u' = \dfrac{1}{2eu}x^{\frac{1}{e}-1} = \dfrac{u^2+1}{2exu}$. It follows

$$\int_1^\infty \frac{1}{x\sqrt{\sqrt[e]{x}-1}}\,\mathrm{d}x = 2e\int_0^\infty \frac{1}{1+u^2}\,\mathrm{d}u = e\pi.$$

   :)

2.2 Fight 2

1. $\displaystyle\int \frac{x^2-1}{x^3+x}\ln\left(x^3+\frac{1}{x^3}+3\left(x+\frac{1}{x}\right)\right)\,\mathrm{d}x$

   Solution. First,

   $$\int \frac{x^2-1}{x^3+x}\ln\left(x^3+\frac{1}{x^3}+3\left(x+\frac{1}{x}\right)\right)\,\mathrm{d}x = 3\int \frac{x^2-1}{x^3+x}\ln\left(x+\frac{1}{x}\right)\,\mathrm{d}x.$$

   Let $u = \ln\left(x+\frac{1}{x}\right)$. Then $\mathrm{d}u = \dfrac{1-\frac{1}{x^2}}{x+\frac{1}{x}}\,\mathrm{d}x = \dfrac{x^2-1}{x^3+x}\,\mathrm{d}x$. So

   $$3\int \frac{x^2-1}{x^3+x}\ln\left(x+\frac{1}{x}\right)\,\mathrm{d}x = 3\int u\,\mathrm{d}u = \frac{3}{2}\left(\ln\left(x+\frac{1}{x}\right)\right)^2 + C.$$

   :)

2. $\displaystyle\int_0^1 \frac{\ln(x)(2x-1)}{\sqrt{x-x^2}}\,\mathrm{d}x$

   Solution. First integrate by parts as

   $$\begin{aligned}\int_0^1 \frac{\ln(x)(2x-1)}{\sqrt{x-x^2}}\,\mathrm{d}x &= \left[-2\ln(x)\sqrt{x-x^2}\right]_0^1 + 2\int_0^1 \frac{\sqrt{x-x^2}}{x}\,\mathrm{d}x \\ &= 2\int_0^1 \frac{\sqrt{x-x^2}}{x}\,\mathrm{d}x.\end{aligned}$$

   This integral can be evaluated by $x = \sin^2(u)$ as

   $$\begin{aligned}2\int_0^1 \frac{\sqrt{x-x^2}}{x}\,\mathrm{d}x &= 2\int_0^{\frac{\pi}{2}} \frac{\sin(u)\cos(u)}{\sin^2(u)}\cdot 2\sin(u)\cos(u)\,\mathrm{d}u \\ &= 4\int_0^{\frac{\pi}{2}} \cos^2(u)\,\mathrm{d}u \\ &= \pi.\end{aligned}$$

   Alternatively, one can perform King's rule before the trigonometric step to avoid integrating $\cos^2(u)$. Or King's rule on $\cos^2(u)$ and add them together. :)

3. $\displaystyle\int_{-2024}^{2025} f_\infty(x)\,\mathrm{d}x$, where $f_n(x) = \begin{cases}\dfrac{2x}{1+x^2}, & n=0;\\ f_{n-1}(f_{n-1}(x)), & n\geq 1, n\in\mathbb{Z};\\ \lim\limits_{k\to\infty} f_k(x), & n=\infty\end{cases}$

   Solution. $f(x) = \lim\limits_{n\to\infty} f_n(x)$ satisfies

   $$f(x) = \frac{2f(x)}{1+f(x)^2} \implies f(x)\in\{-1,0,1\}.$$

   Now observing the signs of f_0, it is clear that $f(x) = \operatorname{sgn}(x)$ (and $f(0)=0$). Then $\displaystyle\int_{-2024}^{2025} f(x)\,\mathrm{d}x = 1$. :)

2.3 Fight 3

1. $\displaystyle\int_3^4 \frac{x-\sqrt{7x-x^2}}{2x-7}\,\mathrm{d}x$

   Solution. By King's rule, we have

   $$I := \int_3^4 \frac{x-\sqrt{7x-x^2}}{2x-7}\,\mathrm{d}x = \int_3^4 \frac{(7-x)-\sqrt{7x-x^2}}{7-2x}\,\mathrm{d}x,$$

   so that

   $$2I = \int_3^4 \frac{x-\sqrt{7x-x^2}-(7-x)+\sqrt{7x-x^2}}{2x-7}\,\mathrm{d}x = \int_3^4 1\,\mathrm{d}x = 1,$$

   and hence $I = \dfrac{1}{2}$. :)

2. $\displaystyle\int_1^\infty \frac{1+2x\ln(2)}{x\sqrt{x4^x-1}}\,\mathrm{d}x$

   Solution. Let $u = x4^x - 1$. Observe that $u' = 4^x(1+2x\ln(2))$. Then we have

   $$\begin{aligned}\int_1^\infty \frac{1+2x\ln(2)}{x\sqrt{x4^x-1}}\,\mathrm{d}x &= \int_1^\infty \frac{4^x(1+2x\ln(2))}{x4^x\sqrt{x4^x-1}}\,\mathrm{d}x\\ &= \int_3^\infty \frac{1}{(u+1)\sqrt{u}}\,\mathrm{d}u\\ &= \int_{\sqrt{3}}^\infty \frac{2}{y^2+1}\,\mathrm{d}y\\ &= 2[\tan^{-1}(y)]_{\sqrt{3}}^\infty\\ &= \frac{\pi}{3}.\end{aligned}$$

   :)

3. $\displaystyle\int \sec(x)(\sec(x)+\tan(x))^{2025}\,\mathrm{d}x$

   Solution. Let $u = \sec(x)+\tan(x)$. Then $u' = \sec(x)\tan(x)+\sec^2(x) = \sec(x)(\sec(x)+\tan(x))$, so that

   $$\int \sec(x)(\sec(x)+\tan(x))^{2025}\,\mathrm{d}x = \int u^{2024}\,\mathrm{d}x = \frac{(\sec(x)+\tan(x))^{2025}}{2025}+C.$$

   :)

2.4 Tiebreaker

1. $\displaystyle\int \frac{e^{e^x}+1}{e^{e^x-x}+1}\,\mathrm{d}x$

   Solution.

   $$\int \frac{e^{e^x}+1}{e^{e^x-x}+1}\,\mathrm{d}x = \int \frac{e^xe^{e^x}+e^x}{e^{e^x}+e^x}\,\mathrm{d}x = \ln\left(e^{e^x}+e^x\right)+C.$$

   :)

3 Battle 3

3.1 Fight 1

1. $\displaystyle\int \frac{\ln(x)+2}{(xe)^{-x}+2}\,\mathrm{d}x$

   Solution. Manipulating a bit, we have

   $$\int \frac{\ln(x)+2}{(xe)^{-x}+2}\,\mathrm{d}x = \int \frac{(xe)^x\ln(x)+2(xe)^x}{1+2(xe)^x}\,\mathrm{d}x.$$

   Now note that $[(xe)^x]' = 2(xe)^x + (xe)^x\ln(x)$. Hence the integral above evaluates to $\dfrac{1}{2}\ln(1+2(xe)^x)+C$. :)

2. $\displaystyle\int_0^\infty \frac{\tan^{-1}\left(\frac{x}{1+2x^2}\right)-\tan^{-1}\left(\frac{x}{1+12x^2}\right)}{x}\,\mathrm{d}x$

   Solution. Using inverse tangent addition formula,

   $$\begin{aligned}
   &\int_0^\infty \frac{\tan^{-1}\left(\frac{x}{1+2x^2}\right)-\tan^{-1}\left(\frac{x}{1+12x^2}\right)}{x}\,\mathrm{d}x\\
   &= \int_0^\infty \frac{\tan^{-1}\left(\frac{2x-x}{1+2x^2}\right)-\tan^{-1}\left(\frac{4x-3x}{1+12x^2}\right)}{x}\,\mathrm{d}x\\
   &= \int_0^\infty \frac{\tan^{-1}(2x)-\tan(x)}{x}\,\mathrm{d}x - \int_0^\infty \frac{\tan^{-1}(4x)-\tan^{-1}(3x)}{x}\,\mathrm{d}x\\
   &= \frac{\pi}{2}\left(\ln(2)-\ln\left(\frac{4}{3}\right)\right)\\
   &= \frac{\pi}{2}(\ln(3)-\ln(2)).
   \end{aligned}$$

   :)

3. $\displaystyle\int_0^{\frac{\pi}{2}} \sin(x)\sqrt{\sin(2x)}\,\mathrm{d}x$

   Solution. By King's rule we have

   $$I := \int_0^{\frac{\pi}{2}} \sin(x)\sqrt{\sin(2x)}\,\mathrm{d}x = \int_0^{\frac{\pi}{2}} \cos(x)\sqrt{\sin(2x)}\,\mathrm{d}x,$$

   so that

   $$\begin{aligned}
   I &= \frac{1}{2}\int_0^{\frac{\pi}{2}} (\sin(x)+\cos(x))\sqrt{\sin(2x)}\,\mathrm{d}x\\
   &= \frac{1}{2}\int_0^{\frac{\pi}{2}} (\sin(x)+\cos(x))\sqrt{1-(\sin(x)-\cos(x))^2}\,\mathrm{d}x.
   \end{aligned}$$

   Let $y = \sin(x)-\cos(x)$, then

   $$I = \frac{1}{2}\int_{-1}^{1} \sqrt{1-y^2}\,\mathrm{d}y = \frac{\pi}{4}.$$

   :)

3.2 Fight 2

1. $\displaystyle\int_{-\infty}^{\infty} \frac{2^{\lfloor 2^x \rfloor + x}}{3^{2^{\lfloor 2^x \rfloor}} + 1}\, \mathrm{d}x$

   Solution. Rewrite as

   $$\int_{-\infty}^{\infty} \frac{2^{\lfloor 2^x \rfloor + x}}{3^{2^{\lfloor 2^x \rfloor}} + 1}\, \mathrm{d}x = \int_{-\infty}^{\infty} \frac{2^x 2^{\lfloor 2^x \rfloor}}{3^{2^{\lfloor 2^x \rfloor}} + 1}\, \mathrm{d}x.$$

   Let $y = 2^x$. Then $dy = y \ln(2)\, \mathrm{d}x$. The integral becomes

   $$\begin{aligned}\frac{1}{\ln(2)} \int_0^{\infty} \frac{2^{\lfloor y \rfloor}}{3^{2^{\lfloor y \rfloor}} + 1}\, \mathrm{d}y &= \frac{1}{\ln(2)} \sum_{n=0}^{\infty} \frac{2^n}{3^{2^n} + 1} \\ &= \frac{1}{\ln(2)} \sum_{n=0}^{\infty} \left(\frac{2^n}{3^{2^n} - 1} - \frac{2^{n+1}}{3^{2^{n+1}} - 1} \right) \\ &= \frac{1}{2\ln(2)}.\end{aligned}$$

   :)

2. $\displaystyle\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \sin(x)^{\tan(x)} \left(\sec^2(x) \ln(\sin(x)) + 1 \right)\, \mathrm{d}x$

   Solution.

   $$\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \sin(x)^{\tan(x)} \left(\sec^2(x) \ln(\sin(x)) + 1 \right)\, \mathrm{d}x = \left[(\sin(x))^{\tan(x)} \right]_{\frac{\pi}{4}}^{\frac{\pi}{2}} = 1 - \frac{\sqrt{2}}{2}.$$

   :)

3. $\displaystyle\int_1^{\sqrt{2}} \frac{\sqrt{x^2 + 2\sqrt{x^2 - 1}}}{x}\, \mathrm{d}x$

   Solution. Let $x = \sec(u)$. Then we have

   $$\begin{aligned}\int_1^{\sqrt{2}} \frac{\sqrt{x^2 + 2\sqrt{x^2 - 1}}}{x}\, \mathrm{d}x &= \int_0^{\frac{\pi}{4}} \frac{\sqrt{\sec^2(u) + 2\tan(u)}}{\sec(u)} \cdot \sec(u)\tan(u)\, \mathrm{d}u \\ &= \int_0^{\frac{\pi}{4}} \tan(u) \sqrt{1 + \tan^2(u) + 2\tan(u)}\, \mathrm{d}u \\ &= \int_0^{\frac{\pi}{4}} \tan(u) + \tan^2(u)\, \mathrm{d}u \\ &= \left[-\ln|\cos(u)| + \tan(u) - u \right]_0^{\frac{\pi}{4}} \\ &= -\ln\left(\frac{\sqrt{2}}{2} \right) + 1 - \frac{\pi}{4} \\ &= \frac{1}{2}\ln(2) + 1 - \frac{\pi}{4}.\end{aligned}$$

   :)

3.3 Fight 3

1. $\displaystyle\int_0^{\infty} \cos\left(x^2 + \frac{1}{x^2} \right)\, \mathrm{d}x$

Solution. By Glassers master theorem,

$$\begin{aligned}\int_0^\infty \cos\left(x^2+\frac{1}{x^2}\right)\,\mathrm{d}x &= \frac{1}{2}\int_{-\infty}^\infty \cos\left(x^2+\frac{1}{x^2}\right)\,\mathrm{d}x\\ &= \frac{1}{2}\int_{-\infty}^\infty \cos(x^2+2)\,\mathrm{d}x\\ &= \int_0^\infty \cos(2)\cos(x^2)-\sin(2)\sin(x^2)\,\mathrm{d}x\\ &= \frac{\sqrt{2\pi}}{4}(\cos(2)-\sin(2)),\end{aligned}$$

where in the last step we have evaluated using the Fresnel integrals. If one does not know Fresnel integrals, the final step can also be evaluated in various ways, for instance converting into exponential form then using the Gaussian integral. :)

2. $\displaystyle\int_0^1 x^e(e^x+e^{-x})+x^{e-1}(e^{x+1}-e^{1-x})\,\mathrm{d}x$

Solution. A double reverse product rule problem. We have

$$\begin{aligned}&\int_0^1 x^e(e^x+e^{-x})+x^{e-1}(e^{x+1}-e^{1-x}))\,\mathrm{d}x\\ &= \int_0^1 x^e e^x + x^{e-1}e^{x+1} + x^e e^{-x} - x^{e-1}e^{-x+1}\,\mathrm{d}x\\ &= [x^e e^x - x^e e^{-x}]_0^1\\ &= e-\frac{1}{e}.\end{aligned}$$

:)

3. $\displaystyle\int \frac{1}{\sin(x)(\ln(\sin(x))-\ln(1+\cos(x)))}\,\mathrm{d}x$

Solution. Let $u=\ln(\sin(x))-\ln(1+\cos(x))$. Then we have

$$u' = \frac{\cos(x)}{\sin(x)} - \frac{-\sin(x)}{1+\cos(x)} = \frac{\cos(x)+\cos^2(x)+\sin^2(x)}{\sin(x)(1+\cos(x))} = \frac{1}{\sin(x)}.$$

Hence we have

$$\begin{aligned}&\int \frac{1}{\sin(x)(\ln(\sin(x))-\ln(1+\cos(x)))}\,\mathrm{d}x\\ &= \int \frac{1}{u}\,\mathrm{d}u\\ &= \ln|\ln(\sin(x))-\ln(1+\cos(x))|+C.\end{aligned}$$

Weierstrass works too. But can you do that in four minutes? :)

3.4 Tiebreaker

1. $\displaystyle\int \frac{1}{(x+2)\sqrt{(x+1)(x+3)}}\,\mathrm{d}x$

Solution. For clarity let $u = x + 2$, then

$$\begin{aligned}\int \frac{1}{(x+2)\sqrt{(x+1)(x+3)}}\,\mathrm{d}x &= \int \frac{1}{u\sqrt{u^2-1}}\,\mathrm{d}u \\ &= \sec^{-1}(u) + C \\ &= \sec^{-1}(x+2) + C.\end{aligned}$$

:)

4 Battle 4

4.1 Fight 1

1. $\displaystyle\int_0^\infty \frac{3^{-\lfloor x \rfloor}\tan(3^{-\lfloor x \rfloor})}{3-\tan^2(3^{-\lfloor x \rfloor})}\,\mathrm{d}x$

   Solution. Using the identity $\tan(3x) = \dfrac{3\tan(x)-\tan^3(x)}{1-3\tan^2(x)}$ gives

   $$\begin{aligned}\frac{\tan(x)}{3-\tan^2(x)} &= \frac{\tan^2(x)}{3\tan(x)-\tan^3(x)}\\ &= \frac{\frac{1}{\tan(x)} - \frac{3-9\tan^2(x)}{3\tan(x)-\tan^3(x)}}{8}\\ &= \frac{1}{8}\left(\cot(x) - 3\cot(3x)\right),\end{aligned}$$

   so that

   $$\begin{aligned}\int_0^\infty \frac{3^{-\lfloor x \rfloor}\tan(3^{-\lfloor x \rfloor})}{3-\tan^2(3^{-\lfloor x \rfloor})}\,\mathrm{d}x &= \sum_{n=0}^{\infty}\frac{3^{-n}\tan(3^{-n})}{3-\tan^2(3^{-n})}\\ &= \frac{1}{8}\sum_{n=0}^{\infty} 3^{-n}\cot(3^{-n}) - 3^{-(n-1)}\cot(3^{-(n-1)})\\ &= \frac{1}{8}\left(\lim_{n\to\infty}\left(3^{-n}\cot(3^{-n})\right) - 3\cot(3)\right)\\ &= \frac{1}{8}\left(1-3\cot(3)\right).\end{aligned}$$

   :)

2. $\displaystyle\int_0^1 x^3\sin^2(\ln(x))\cos^2(\ln(x))\,\mathrm{d}x$

   Solution. We have

   $$\sin^2(\ln(x))\cos^2(\ln(x)) = \frac{1}{4}\sin^2(2\ln(x)) = \frac{1}{8} - \frac{1}{8}\cos(4\ln(x)).$$

   Let $x = e^y$. Then we have

   $$\begin{aligned}\int_0^1 x^3\sin^2(\ln(x))\cos^2(\ln(x))\,\mathrm{d}x &= \frac{1}{8}\int_0^1 x^3 - \frac{1}{8}\int_0^1 x^3\cos(4\ln(x))\,\mathrm{d}x\\ &= \frac{1}{32} - \frac{1}{8}\operatorname{Re}\left(\int_{-\infty}^0 e^{4y+4yi}\,\mathrm{d}y\right)\\ &= \frac{1}{32} - \frac{1}{8}\operatorname{Re}\left(\left[\frac{e^{4y(1+i)}}{4(1+i)}\right]_{-\infty}^0\right)\\ &= \frac{1}{32} - \frac{1}{32}\operatorname{Re}\left(\frac{1}{1+i}\right)\\ &= \frac{1}{64}.\end{aligned}$$

   :)

3. $\lim_{n\to\infty} n\int_0^{\frac{\pi}{4}} \tan^n(x)\,\mathrm{d}x$

   Solution. Let $I_n = \int_0^{\frac{\pi}{4}} \tan^n(x)\,\mathrm{d}x$. Then we have

$$\begin{aligned} I_n + I_{n+2} &= \int_0^{\frac{\pi}{4}} \tan^n(x)(1+\tan^2(x))\,\mathrm{d}x \\ &= \left[\frac{1}{n+1}\tan^{n+1}(x)\right]_0^{\frac{\pi}{4}} \\ &= \frac{1}{n+1} \end{aligned}$$

   Note that I_n is a decreasing function of n since $0 < \tan(x) \leq 1$ in $\left(0, \frac{\pi}{4}\right)$, so that

$$\frac{1}{n+1} = I_n + I_{n+2} \leq 2I_n \leq I_n + I_{n-2} = \frac{1}{n-1}$$

   so that the required limit is $\frac{1}{2}$. :)

4.2 Fight 2

1. $\int_0^{\frac{\pi}{2^m}} \frac{\sin((2^m + 2025(2^n))x)}{\sin(2^m x)} \prod_{k=1}^{m-n} \cos(2^{m-k}x)\,\mathrm{d}x$, $m, n \in \mathbb{Z}$, $m > n$

   Solution. Use sine double angle multiple times. We have

$$\begin{aligned} &\int_0^{\frac{\pi}{2^m}} \frac{\sin((2^m + 2025(2^n))x)}{\sin(2^m x)} \prod_{k=1}^{m-n} \cos(2^{m-k}x)\,\mathrm{d}x \\ &= \int_0^{\frac{\pi}{2^m}} \frac{\sin((2^m + 2025(2^n))x)}{\sin(2^m x)} \cdot \frac{\sin(2^n x)}{\sin(2^n x)} \prod_{k=1}^{m-n} \cos(2^{m-k}x)\,\mathrm{d}x \\ &= \frac{1}{2^{m-n}} \int_0^{\frac{\pi}{2^m}} \frac{\sin((2^m + 2025(2^n))x)}{\sin(2^m x)} \cdot \frac{\sin(2^m x)}{\sin(2^n x)}\,\mathrm{d}x \\ &= \frac{1}{2^{m-n}} \int_0^{\frac{\pi}{2^n}} \frac{\sin((2^m + 2025(2^n))x)}{\sin(2^n x)}\,\mathrm{d}x \\ &= \frac{1}{2^{m-2n}} \int_0^{\pi} \frac{\sin((2^{m-n} + 2025)y)}{\sin(y)}\,\mathrm{d}y \\ &= \frac{\pi}{2^{m-2n}}, \end{aligned}$$

   where in the last line we have used the famous evaluation

$$\int_0^{\pi} \frac{\sin(nx)}{\sin(x)}\,\mathrm{d}x = \begin{cases} 0, & n \text{ even;} \\ \pi, & n \text{ odd.} \end{cases}$$

   :)

2. $\int \frac{e^{W(x)}}{x^2}\,\mathrm{d}x$, where W is the Lambert W function on the principal branch (i.e. the unique solution w to $we^w = x$ on the principal branch)

   Note: Answers in terms of $W(x)$ are acceptable

Solution. We have $W'(x) = \dfrac{1}{W(x)e^{W(x)} + e^{W(x)}}$. Then, writing $w = W(x)$, we have

$$\begin{aligned}\int \frac{e^{W(x)}}{x^2}\,\mathrm{d}x &= \int \frac{e^w(we^w + e^w)}{w^2e^{2w}}\,\mathrm{d}w \\ &= \int \frac{w+1}{w^2}\,\mathrm{d}w \\ &= \ln|w| - \frac{1}{w} + C \\ &= \ln|W(x)| - \frac{1}{W(x)} + C.\end{aligned}$$

:)

3. $\displaystyle\int_{-\infty}^{\infty} \frac{\sin^2(x)}{45x^2 + x^2\tan^{-1}(x) + x^2\sqrt{2025 + (\tan^{-1}(x))^2}}\,\mathrm{d}x$

Solution. For $e(x)$ and $o(x)$ even and odd functions respectively, we have

$$\begin{aligned}I &:= \int_{-a}^{a} \frac{e(x)}{b + o(x) + \sqrt{b^2 + o(x)^2}}\,\mathrm{d}x \\ &= \int_{-a}^{a} \frac{e(-x)}{b + o(-x) + \sqrt{b^2 + o(-x)^2}}\,\mathrm{d}x \\ &= \int_{-a}^{a} \frac{e(x)}{b - o(x) + \sqrt{b^2 + o(x)^2}}\,\mathrm{d}x,\end{aligned}$$

so that

$$\begin{aligned}I &= \frac{1}{2}\int_{-a}^{a} \frac{e(x)}{b + o(x) + \sqrt{b^2 + o(x)^2}} + \frac{e(x)}{b - o(x) + \sqrt{b^2 + o(x)^2}}\,\mathrm{d}x \\ &= \frac{1}{2b}\int_{-a}^{a} e(x)\,\mathrm{d}x \\ &= \frac{1}{b}\int_{0}^{a} e(x)\,\mathrm{d}x.\end{aligned}$$

Now in the problem, the integrand is even, so we have

$$\begin{aligned}&\int_{\infty}^{\infty} \frac{\sin^2(x)}{45x^2 + x^2\tan^{-1}(x) + x^2\sqrt{2025 + (\tan^{-1}(x))^2}}\,\mathrm{d}x \\ &= \frac{1}{45}\int_{0}^{\infty} \frac{\sin^2(x)}{x^2}\,\mathrm{d}x \\ &= \frac{\pi}{90}.\end{aligned}$$

:)

4.3 Fight 3

1. $\displaystyle\int_0^{\frac{\pi}{4}} \frac{1}{\sqrt[4]{2} + \sqrt{1 + \tan(x)}}\,\mathrm{d}x$

Solution. Secretly a King's rule problem, even if it doesn't look it. Let $u = \frac{\pi}{4} - x$. Then we have

$$\begin{aligned} I &:= \int_0^{\frac{\pi}{4}} \frac{1}{\sqrt[4]{2} + \sqrt{1+\tan(x)}}\,\mathrm{d}x \\ &= \int_0^{\frac{\pi}{4}} \frac{1}{\sqrt[4]{2} + \sqrt{1+\tan\left(\frac{\pi}{4}-u\right)}}\,\mathrm{d}u \\ &= \int_0^{\frac{\pi}{4}} \frac{1}{\sqrt[4]{2} + \sqrt{1+\frac{1-\tan(u)}{1+\tan(u)}}}\,\mathrm{d}u \\ &= \int_0^{\frac{\pi}{4}} \frac{1}{\sqrt[4]{2} + \sqrt{\frac{2}{1+\tan(u)}}}\,\mathrm{d}u \\ &= \int_0^{\frac{\pi}{4}} \frac{\frac{\sqrt{1+\tan(u)}}{\sqrt[4]{2}}}{\sqrt{1+\tan(u)} + \sqrt[4]{2}}\,\mathrm{d}u. \end{aligned}$$

Hence,

$$2I = \int_0^{\frac{\pi}{4}} \frac{1 + \frac{\sqrt{1+\tan(x)}}{\sqrt[4]{2}}}{\sqrt{1+\tan(x)} + \sqrt[4]{2}}\,\mathrm{d}x = \int_0^{\frac{\pi}{4}} \frac{1}{\sqrt[4]{2}}\,\mathrm{d}x = \frac{\pi}{4\sqrt[4]{2}},$$

so that $I = \frac{\pi}{8\sqrt[4]{2}}$. :)

2. $\displaystyle\int_0^{\infty} \frac{\lfloor x \rfloor !(2\lfloor x \rfloor)!!}{(2\lceil x \rceil - 1)!}\,\mathrm{d}x$

Solution. Rewrite it in the form

$$\int_0^{\infty} \frac{\lfloor x \rfloor !(2\lfloor x \rfloor)!!}{(2\lceil x \rceil - 1)!}\,\mathrm{d}x = \sum_{n=0}^{\infty} \frac{n!(2n)!!}{(2n+1)!} = \sum_{n=0}^{\infty} \frac{n!n!2^n}{(2n+1)!}.$$

Now recall the beta function representation

$$B(x,y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)} \implies B(n+1,n+1) = \frac{n!n!}{(2n+1)!}.$$

Then the sum is

$$\begin{aligned} \sum_{n=0}^{\infty} 2^n B(n+1,n+1) &= \sum_{n=0}^{\infty} 2^n \int_0^1 x^k(1-x)^k\,\mathrm{d}x \\ &= \int_0^1 \sum_{n=0}^{\infty} 2^n x^k (1-x)^k\,\mathrm{d}x \\ &= \int_0^1 \frac{1}{1-2x(1-x)}\,\mathrm{d}x \\ &= \int_0^1 \frac{2}{1+(2x-1)^2}\,\mathrm{d}x \\ &= \left[\tan^{-1}(2x-1)\right]_0^1 \\ &= \frac{\pi}{2}. \end{aligned}$$

:)

3. $\displaystyle\int_1^3 \frac{\ln(x)}{x^2+3}\,\mathrm{d}x$

Solution. Split the range as

$$I := \int_1^3 \frac{\ln(x)}{x^2+3}\,\mathrm{d}x = \int_1^{\sqrt{3}} \frac{\ln(x)}{x^2+3}\,\mathrm{d}x + \int_{\sqrt{3}}^3 \frac{\ln(x)}{x^2+3}\,\mathrm{d}x.$$

In the second integral let $x = \dfrac{3}{u}$, and we get

$$\int_{\sqrt{3}}^3 \frac{\ln(x)}{x^2+3}\,\mathrm{d}x = \int_1^{\sqrt{3}} \frac{\ln\left(\frac{3}{u}\right)}{\left(\frac{3}{u}\right)^2+3}\cdot\frac{3}{u^2}\,\mathrm{d}u = \int_1^{\sqrt{3}} \frac{\ln(3)-\ln(u)}{u^2+3}\,\mathrm{d}u.$$

It follows that

$$\begin{aligned} I &= \int_1^{\sqrt{3}} \frac{\ln(3)}{u^2+3}\,\mathrm{d}u \\ &= \frac{\ln(3)}{\sqrt{3}}\left[\tan^{-1}\left(\frac{x}{\sqrt{3}}\right)\right]_1^{\sqrt{3}} \\ &= \frac{\pi\ln(3)}{12\sqrt{3}}. \end{aligned}$$

:)

4.4 Tiebreaker

1. $\displaystyle\int_{-2024}^{2025} \frac{1}{3^x+\sqrt{3}}\,\mathrm{d}x$

Solution. Observe the identity

$$\begin{aligned} \frac{1}{3^x+\sqrt{3}} + \frac{1}{3^{1-x}+\sqrt{3}} &= \frac{3^{1-x}+\sqrt{3}+3^x+\sqrt{3}}{(3^x+\sqrt{3})(3^{1-x}+\sqrt{3})} \\ &= \frac{2\sqrt{3}+3^{1-x}+3^x}{2\cdot 3+\sqrt{3}\cdot 3^{1-x}+\sqrt{3}\cdot 3^x} \\ &= \frac{1}{\sqrt{3}}, \end{aligned}$$

Hence reflection about $x = \dfrac{1}{2}$ gives

$$\int_{-2024}^{2025} \frac{1}{3^x+\sqrt{3}}\,\mathrm{d}x = \frac{4049}{2\sqrt{3}}.$$

Alternative which provides more motivation: Factor out the $\sqrt{3}$ to get

$$I = \frac{1}{\sqrt{3}}\int_{-2024}^{2025} \frac{\mathrm{d}x}{3^{x-\frac{1}{2}}+1}$$

This has symmetry about $x = 1/2$ clearly so the substitution $u = 1-x$ (King's rule) works immediately, giving

$$I = \frac{1}{\sqrt{3}}\int_{-2024}^{2025} \frac{\mathrm{d}x}{3^{\frac{1}{2}-x}+1}$$

Therefore

$$2I = \frac{1}{\sqrt{3}} \int_{-2024}^{2025} \mathrm{d}x = \frac{4049}{\sqrt{3}} \implies I = \frac{4049}{2\sqrt{3}}.$$

One more way: just get the indefinite. It's a bit cursed that way though. :)